

Dataset description

This heart disease dataset was collected from a multispecialty hospital in India. It includes data from 1,000 individuals and comprises 12 key features commonly associated with heart disease. Although initially described as containing over 14 features, the final dataset includes 12 carefully selected attributes that are highly relevant for research. This dataset is valuable for developing early-stage heart disease detection systems and training predictive machine learning models.

الوصف (Description): بالعربية:

تم جمع مجموعة بيانات أمراض القلب هذه من أحد المستشفيات متعددة التخصصات في الهند. وتضم بيانات 1000 فرد وتشمل 12 سمة رئيسية مرتبطة بشكل شائع بأمراض القلب. وعلى الرغم من أنه تمت الإشارة في البداية إلى أنها تحتوي على أكثر من 14 سمة، فإن النسخة النهائية من البيانات تتضمن 12 سمة مختارة بعناية وذات صلة كبيرة بالبحث العلمي. وتعد هذه المجموعة من البيانات ذات قيمة في تطوير أنظمة للكشف المبكر عن أمراض القلب، وكذلك في تدريب نماذج التنبؤ باستخدام تقنيات تعلم الآلة.

Cardiovascular Disease Dataset Description

No	Attribute	Assigned Code	Unit	Type of the Data
1	Patient Identification Number	patientid	Number	Numeric
2	Age	age	In Years	Numeric
3	Gender	gender	1,0(0= female, 1 = male)	Binary
4	Chest pain type	chestpain	0,1,2,3 (Value 0: typical angina Value 1: atypical angina Value 2: non-anginal pain Value 3: asymptomatic)	Nominal
5	Resting blood pressure	restingBP	94-200 (in mm HG)	Numeric
6	Serum cholesterol	serumcholesterol	126-564 (in mg/dl)	Numeric
7	Fasting blood sugar	fastingbloodsugar	0,1 > 120 mg/dl (0 = false , 1 = true)	Binary
8	Resting electrocardiogram results	restingelectro	0,1,2 (Value 0: normal, Value 1: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV), Value 2: showing probable or definite left ventricular hypertrophy by Estes' criteria)	Nominal
9	Maximum heart rate achieved	maxheartrate	71-202	Numeric
10	Exercise induced angina	exerciseangia	0,1 (0 = no, 1 = yes)	Binary
11	Oldpeak =ST	oldpeak	0-6.2	Numeric
12	Slope of the peak exercise ST segment	slope	1,2,3 (1-upsloping, 2-flat, 3-downsloping)	Nominal
13	Number of major vessels	noofmajorvessels	0,1,2,3	Numeric

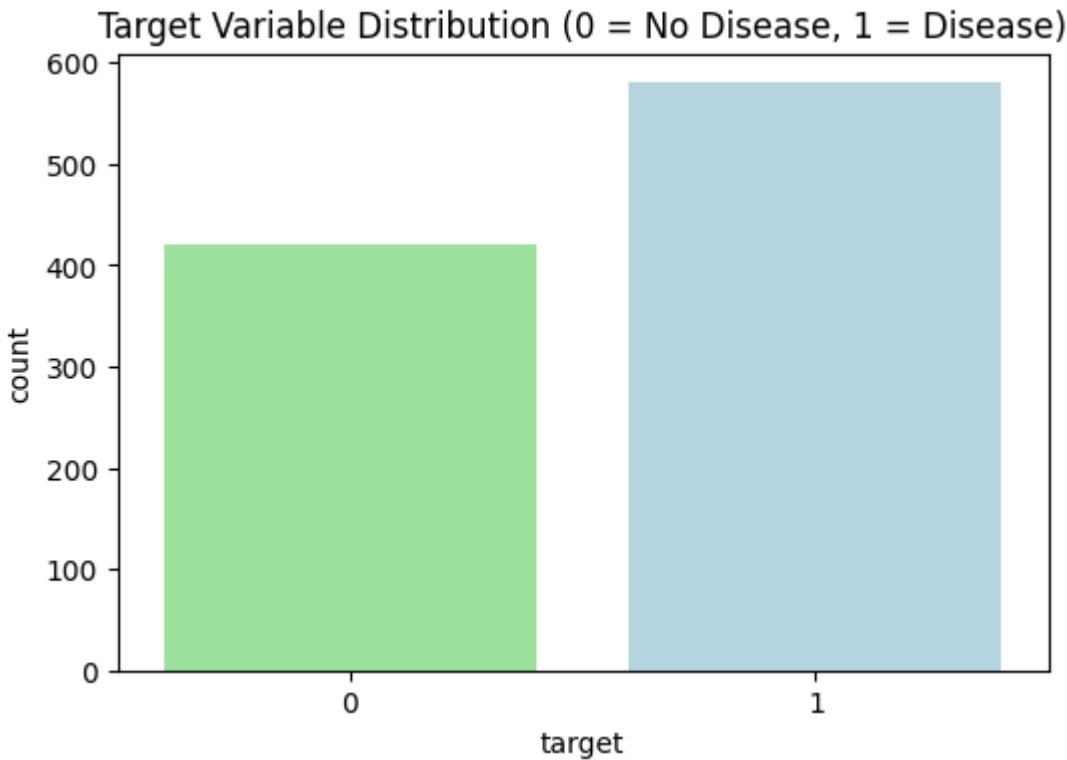
14	Classification	target	0,1 (0= Absence of Heart Disease, 1= Presence of Heart Disease)	Binary
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Essential Python Libraries for Data Analysis and Modeling

In this project, several essential Python libraries were imported to support data analysis, preprocessing, model building, and evaluation. **Pandas** and **NumPy** were used for data manipulation and numerical computations. **Matplotlib** and **Seaborn** helped visualize the dataset and explore relationships between features.

For machine learning, **scikit-learn** provided a comprehensive suite of tools for preprocessing, model training, and performance evaluation, including algorithms like Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machines (SVM). Additionally, **XGBoost** and **Random Forest** from `xgboost` and `sklearn.ensemble` were used for advanced ensemble modeling techniques. These libraries collectively enabled efficient development and analysis of the predictive models.

Target Variable Distribution: Diseased vs. Non-Diseased Cases



The graph shows the distribution of the target variable, where:

- Class 1 (Disease) has more cases than class 0 (No Disease),
- There is a **moderate class imbalance** (more Disease cases),

Data Preprocessing

Before training the machine learning models, several preprocessing steps were applied to ensure data quality and optimize model performance:

- **Handling Missing Values:**
The dataset was examined for missing values. If any were found, strategies such as mean imputation for numerical features and mode imputation for categorical features were used.

```
print("Missing values:\n", df.isnull().sum())
```

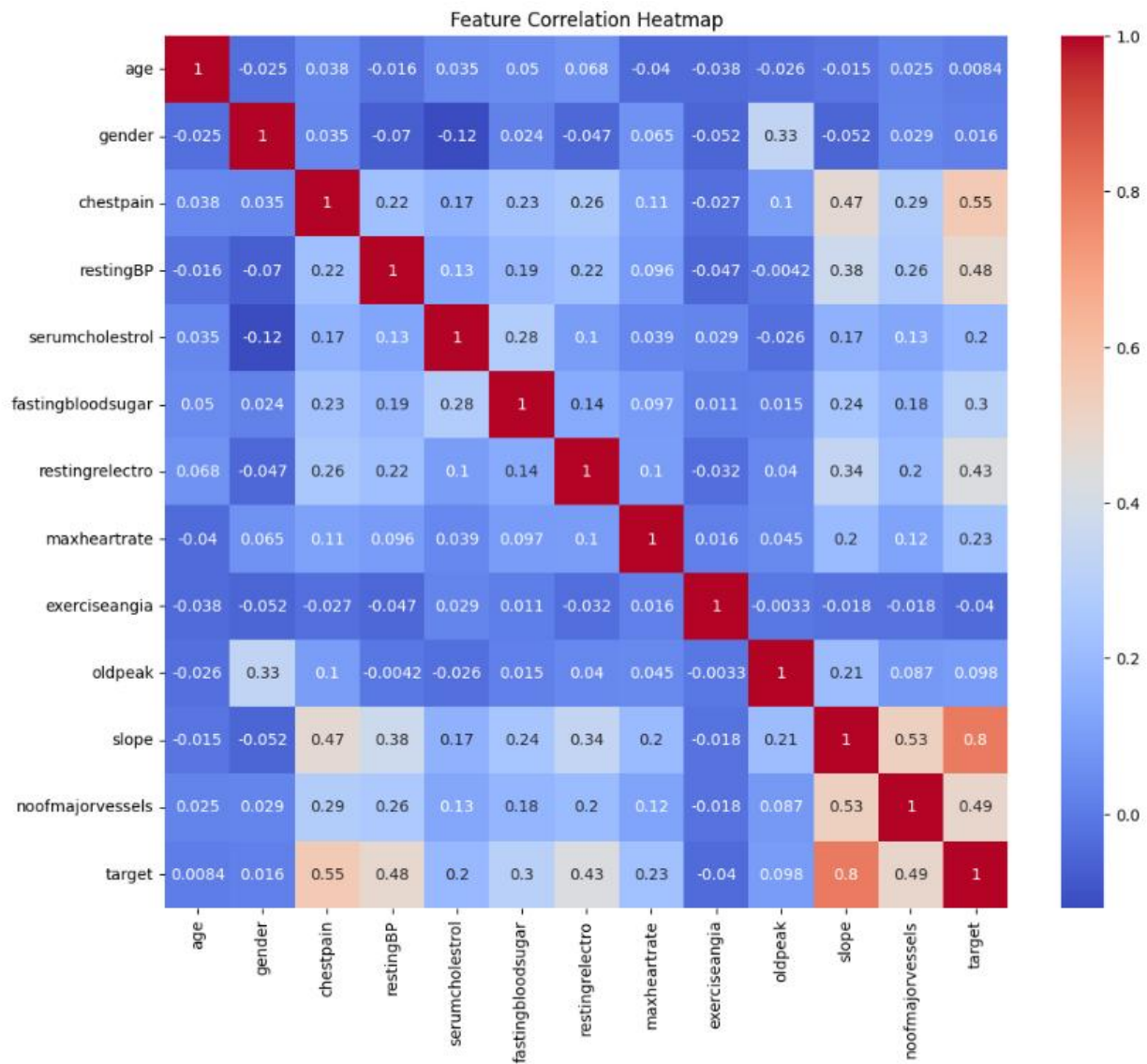
```
Missing values:
age          0
gender       0
chestpain    0
restingBP    0
serumcholesterol  0
fastingbloodsugar  0
restingrelectro  0
maxheartrate  0
exerciseangia  0
oldpeak      0
slope        0
noofmajorvessels  0
target       0
dtype: int64
```

- **Encoding Categorical Variables:**
Categorical features like `gender`, `chestpain`, `restingrelectro`, and `slope` were encoded into numerical formats using either one-hot encoding or label encoding, based on the model's requirements.
- **Feature Scaling:**
Numerical features such as `restingBP`, `serumcholesterol`, `maxheartrate`, and `oldpeak` were scaled using standardization (z-score normalization) to ensure uniformity across features.

```
# Feature scaling
X = df.drop('target', axis=1)
y = df['target']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

- **The Correlation Heatmap**



The correlation heatmap illustrates the relationships between different features in the dataset and their association with the target variable (presence of heart disease).

The heatmap shows that chestpain, slope, restingelectro, and noofmajorvessels have the strongest positive correlations with heart disease.

Among them, slope has the highest correlation (0.8) with the target variable.

Other features like age, gender, and fastingbloodsugar show very weak correlations, indicating lower predictive importance.

No strong multicollinearity was observed between independent variables, allowing safe use of features in modeling.

Data Splitting:

The dataset was divided into a training set (80%) and a testing set (20%) to validate model performance on unseen data.

Model Building

Several machine learning models were developed to predict the risk of heart disease:

- **Logistic Regression (LR):**
A linear model ideal for binary classification, providing a strong baseline.
- **Support Vector Machine (SVM):**
A robust classifier that finds the optimal hyperplane separating the two classes.
- **Random Forest (RF):**
An ensemble learning method that builds multiple decision trees and aggregates their results.
- **Extreme Gradient Boosting (XGBoost):**
A highly effective boosting technique designed to improve performance on structured data.

Model Evaluation



```
Model: Logistic Regression
Classification Report:
              precision    recall  f1-score   support

     0       0.96      0.95      0.96         83
     1       0.97      0.97      0.97        117

 accuracy          0.96          200
 macro avg         0.96          200
 weighted avg      0.96          200
```

```
Model: Support Vector Machine
Classification Report:
              precision    recall  f1-score   support

     0       0.98      0.98      0.98         83
     1       0.98      0.98      0.98        117

 accuracy          0.98          200
 macro avg         0.98          200
 weighted avg      0.98          200
```

```
Model: Random Forest
Classification Report:
              precision    recall  f1-score   support

     0       1.00      0.98      0.99         83
     1       0.98      1.00      0.99        117

 accuracy          0.99          200
 macro avg         0.99          200
 weighted avg      0.99          200
```

```

Model: XGBoost
Classification Report:
              precision    recall  f1-score   support

     0       0.96       0.95       0.96         83
     1       0.97       0.97       0.97        117

 accuracy      0.96
 macro avg     0.96
 weighted avg  0.96

```

Model Performance Comparison

```

# Step 9: Compare model performance
results_df = pd.DataFrame(results, index=['Accuracy', 'Precision', 'Recall', 'F1-Score']).T
print("\nModel Performance Comparison:")
display(results_df)

```



Model Performance Comparison:

	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.965	0.966102	0.974359	0.970213
Support Vector Machine	0.980	0.982906	0.982906	0.982906
Random Forest	0.975	0.974576	0.982906	0.978723
XGBoost	0.965	0.966102	0.974359	0.970213



The model performance comparison shows that all machine learning models achieved high accuracy, precision, recall, and F1-scores.

- **The Support Vector Machine (SVM)** performs the best across all metrics, achieving the highest values in accuracy (0.980), Precision (0.9829), Recall (0.9829), and F1-Score (0.9829).

- **Random Forest** follows closely, with strong performance in Precision (0.9746), Recall (0.9829), and F1-Score (0.9787), while its Accuracy is slightly lower (0.975).

- **Logistic Regression** and **XGBoost** have identical performance, both showing an Accuracy of 0.965, Precision of 0.9661, Recall of 0.9744, and F1-Score of 0.9702.

Overall, the SVM model is the top performer, with Random Forest being a strong alternative for this particular dataset.

Interactive Interfaces for our model:

We use the Gradio open-source Python library, which allows developers to quickly create interactive user interfaces for machine learning models. Gradio enables users to effortlessly build web-based interfaces for testing, sharing, and deploying models in a user-friendly way—without the need for complex frontend development skills.

Heart Attack Risk Predictor (SVM)

Enter patient data to predict the risk of heart attack.

Age	31
Gender (0 = Female, 1 = Male)	1
Chestpain (0 = typical angina, 1 = atypical angina, 2=non-anginal pain, 3= asymptomatic)	1
RestingBP	99
Serumcholesterol	0
Fasting blood sugar, 0,1 > 120 mg/dl. (0 = false , 1 = true)	0
Resting electrocardiogram results (0 = normal, 1 = ST-T wave abnormality, 2=left ventricular hypertrophy)	2
Maximum heart rate achieved	136
Exercise induced angina (0 = No, 1 = yes)	0
oldpeak	

5.3

Slope of the peak exercise ST segment (1 = upsloping, 2 = flat, 3=downsloping)

3

Number of major vessels

2

Clear

Submit

Heart Attack Risk Prediction

High Risk of Heart Attack (Confidence: 0.99)

Flag

The interfaces were tested with new data, and the results were outstanding.

Conclusion:

This project aimed to develop an early detection system for heart disease using a dataset from a multispecialty hospital in India, which included data from 1,000 individuals with 12 key features related to heart disease. Essential Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, XGBoost, and Random Forest were used for data analysis, model building, and evaluation.

Data preprocessing revealed strong correlations between certain features (e.g., chest pain, slope, resting electrocardiographic results) and heart disease. The Support Vector Machine (SVM) outperformed other models, achieving the highest accuracy (0.980), precision (0.9829), recall (0.9829), and F1-Score (0.9829). Random Forest was a strong alternative, while Logistic Regression and XGBoost showed similar performance.

Gradio was used to create interactive web interfaces for the models, enabling easy testing, sharing, and deployment without the need for complex frontend development. The SVM model proved to be the most effective for heart disease detection in this project.